Srinivasan P.

(520) 365-5332 | Tucson, AZ (Open to Relocation) | srinivasan.poonkundran@gmail.com | LinkedIn | GitHub | Portfolio

Professional Experience

Research Assistant — Feb. 2026 – Present
The University of Arizona — *Tucson, AZ*

- Conduct data analysis and implement machine learning models to support ongoing research initiatives in collaboration with faculty and graduate researchers.
- Develop and evaluate computational workflows and experimental pipelines for data-driven studies.
- Contribute to documentation, reproducible research code, and preparation of technical materials.

Software Engineer — Sept. 2021 – Jan. 2024
Cognizant Technology Solutions — *Remote*

- Built Java Spring Boot microservices deployed on AWS to support core business workflows, backed by MySQL databases and integrated with Angular dashboards for KPI visualization used by internal stakeholders.
- Improved backend performance by optimizing SQL queries and caching critical endpoints, reducing average page load times from 2.2s to 1.5s.
- Built a Python-based PDF label validation tool to automate manual checks, reducing operational effort and saving approximately $1.6K annually.
- Contributed to code reviews and refactored backend services with Java and Spring Boot, improving code quality and reducing defect occurrences, which accelerated release cycles.

Software Developer — Jan. 2023 – Jan. 2024
Self-Employed — *Part-time, Remote*

- Developed an Android-based billing and management application supporting invoices, sales, and staff operations for a retail store processing 1,000+ monthly transactions.
- Designed in-app sales dashboards to track daily revenue, top-selling items, and seasonal trends, enabling data-driven inventory and pricing decisions.
- Integrated secure cloud storage using AWS S3 for invoices and records, enabling reliable backup, cross-device access, and recovery from device failures.

Projects

GradeLens – RAG-Powered AI Agent for Intelligent Evaluation | GitHub

- Architected a Retrieval-Augmented Generation system using LangChain and the Anthropic Claude API, integrating a pgvector-based vector database for semantic search and context-aware response generation.
- Designed a tool-enabled AI grading agent using Agno with deterministic rubric validation and iterative retry logic, ensuring structured, schema-compliant grading outputs.
- Implemented asynchronous API handling and parallel execution to grade multiple exam questions simultaneously, significantly reducing overall processing time.
- Integrated the AI pipeline into a full-stack Django + React application to deliver scalable, low-latency user interactions.

Emotion Detection from Text | GitHub

- Built a transformer-based NLP system using BERT embeddings and Bi-GRU layers to classify text into 7 emotions, achieving 0.83 F1-micro score.
- Designed the preprocessing and training pipeline using Hugging Face and TensorFlow to handle large-scale text data efficiently.
- Optimized model performance through structured tuning and regularization to improve generalization.

Education

The University of Arizona — Jan. 2024 – Dec. 2025
Master of Science, Data Science — *Tucson, AZ*
Honors: Distinguished Graduate Scholar (Fall 2025)

Savitribai Phule Pune University — Aug. 2017 – May 2021
Bachelor of Engineering, Computer Engineering — *Pune, India*

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, R, C, C++
Web Technologies: HTML, CSS, Node.js, Angular, React, Spring Boot, Spring MVC, Django REST Framework, RESTful Services, JWT
Development Practices: Object Oriented Programming, Microservice Architecture, Data Structures, CI/CD, Maven
Database & Cloud: MySQL, MongoDB, AWS - EC2, ECS, Load Balancers, Elastic Beanstalk, S3, IAM
AI Systems: Agno, LangChain, Retrieval-Augmented Generation (RAG), Vector Databases (pgvector)
Certifications: Amazon Web Services Certified Cloud Practitioner | Credentials